

Project Standard

Rubric items: 6

Item 1: 6 points

The cover page must be presented with the title and the date of completion (+2 pts.). A complete problem description must be given in Chapter 1 (+4 pts.). Deduct points if:

- the cover page is not complete (-1)
- the introduction is a simple copy + paste of the assignment statement (-3)
- the introduction is not very clear -- in this chapter one is supposed to make it clear **WHAT is to be done, besides why one is doing it** (-1 ~ -2)
- others - please specify in the final comments.

Item 2: 12 points

Chapter 2 is supposed to contain the descriptions (**pseudo-code** preferred) of all the key algorithms involved (+3 pts. for the data structures; +7 pts. for the algorithms), plus a **sketch of the main program** (+2 pts.). Deduct points if:

- the algorithm specification is not complete – the data structure description is missing (-2)
- the algorithm specification is not complete – the key algorithm is missing (-3 ~ -7)
- not making one's algorithm easier to understand than a simple program (-2)
- only a program + comments, which is not acceptable (-4)
- others - please specify in the final comments.

Item 3: 2 points

The overall style of documentation is supposed to be neat and clear. Deduct (at most 2) points if:

- the document looks messy - some of the data in the charts and tables are missing (-1)
- the hand-in is not zipped with proper folders (-1)
- the hand-in is not complete - some files are missing (-1)
- others - please specify in the final comments.

Item 4: 20 points

A complete table of test cases with testing purposes must be given in Chapter 3. A minimum of 3 test cases must be given (+10 pts.). Besides **at least one comprehensive test** (+6 pts.), the cases with **the smallest or the largest sizes**, and some **extreme cases must be covered** (+4 pts.). Deduct points if:

- the testing results contain some test cases, however with no **specification** on their purposes (-3)
- the testing results contain some test cases, but there are still bugs missed (-1 ~ -10)
- the testing results contain too few cases and hence is too simple to be considered as being complete (-4 ~ -10)
- others - please specify in the final comments.

Item 5: 10 points

Analysis of both the time (+5 pts.) and space (+5 pts.) complexities of the algorithms are supposed to be given in Chapter 4. Bonus point for **discussing more than just one algorithm** (+2 pts.). Deduct points if:

- analysis of the complexities of time and/or space is missing -- one must show how one has reached the conclusions, instead of simply listing them (-4)
- others - please specify in the final comments.

Item 6: 50 points

For the programming work, deduct points if:

- the programs are not working properly (-1 ~ -20)
- the programs are not **well-commented** (-50)
- the **programming style** is too messy to be judged (-1 ~ -5)
- others - please specify in the final comments.